

CHS GEOMETRIC IMPERFECTION MODELLING

Analytical formulation • Abaqus automation • nonlinear FE • dataset generation • machine learning • 3D-printed materials

RESEARCH Internship

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01

Background & motivation

Why geometric imperfections matter for nonlinear thin-walled structural response.

01 | Why imperfection modelling matters

Imperfections affect the starting geometry, buckling trigger and nonlinear equilibrium path.

Thin-walled efficiency

Mass-efficient shells are susceptible to local and interactive buckling.

Imperfection sensitivity

Small initial deviations can alter instability initiation and post-buckling response.

Mode interaction

Geometric and material nonlinearity can amplify sensitivity.

Engineering need

Imperfections must be defined by shape, amplitude and spatial wavelength.

INSERT: PERFECT CHS → IMPERFECT CHS → BUCKLED CHS



02

Literature & research gap

The project adapts the mechanics of the supplied Shen–Wadee study to a CHS computational framework.

02 | What Shen–Wadee establishes

The supplied 2019 paper provides the conceptual foundation.

KEY FINDING

The lowest-load local eigenmode does not necessarily represent the most severe imperfection profile.

PERIODIC FORM

Longitudinal wavelength $\Lambda = 2L/\beta$; β is taken as an odd number in the periodic formulation.

NONLINEAR ANALYSIS

Riks arc-length analysis is used to trace nonlinear equilibrium paths.

PROJECT EXTENSION

- Convert the cross-sectional idea to a continuous CHS circumferential function.
- Establish imperfection equation
- Automate nodal coordinate modification using Python.
- Study D/t , A/t , m and β systematically.
- Build a traceable FE database for future ML modelling.



03

Aim & objectives

One integrated pipeline from analytical equation to nonlinear FE and future data-driven prediction.

03 | Project objectives

The objectives are arranged in the order of the actual computational workflow.

01

Formulate a 3D periodic radial imperfection.

02

Use LBA to identify candidate m and β .

03

Automate Abaqus node transformation.

04

Generate imperfect .inp and .csv files.

05

Run nonlinear FE / Riks cases.

06

Automate response extraction.

07

Build a traceable FE dataset.

08

Develop a future ML surrogate.

09

Extend the architecture to 3D-printed materials.



04

CHS geometry & coordinates

The geometry is described in cylindrical coordinates before the node transformation.

04 | Geometry and coordinate system

Mean radius + radial imperfection → Cartesian node coordinates.

MEAN RADIUS

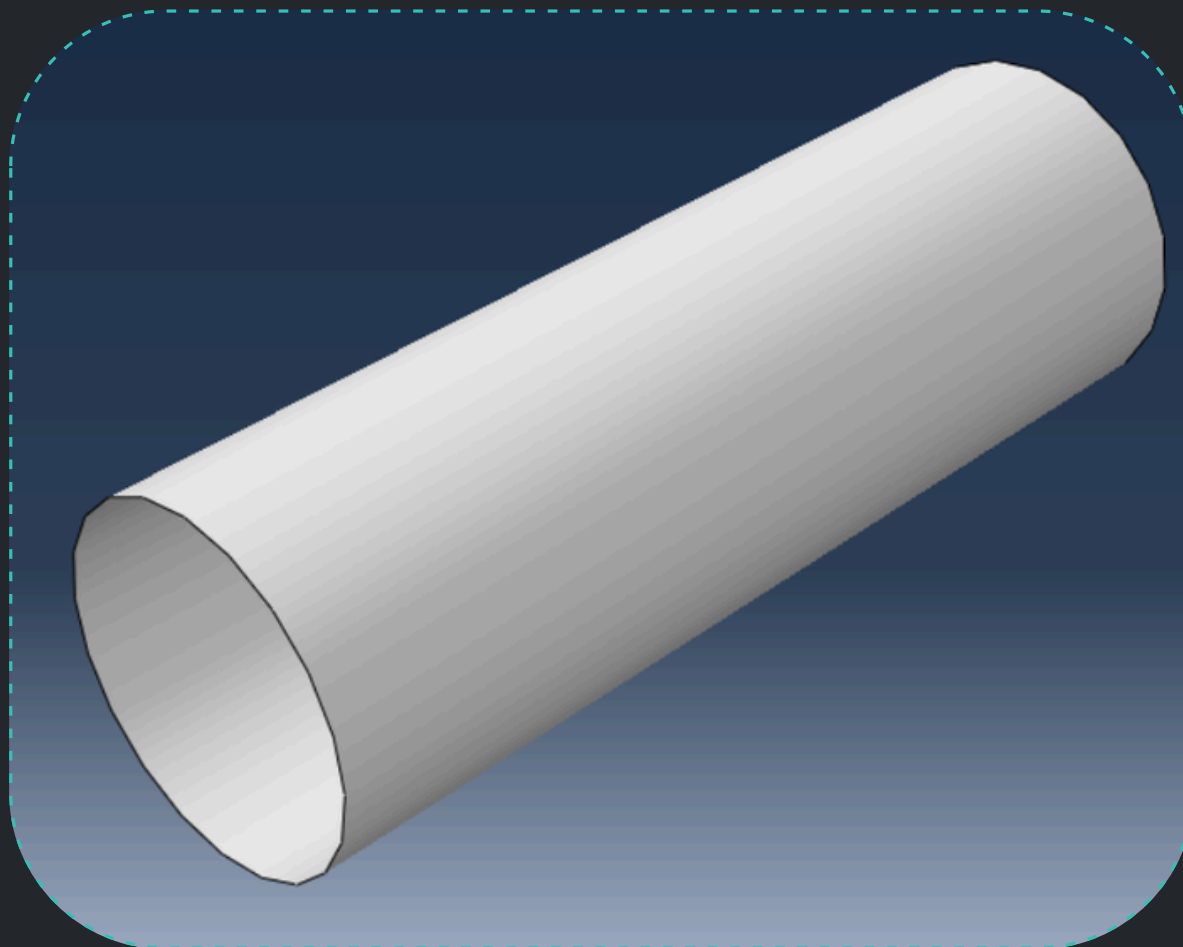
$$R_m = (D - t) / 2$$

IMPERFECT RADIUS

$$r_{\text{imp}} = R_m + \Delta r(\theta, z)$$

CARTESIAN

$$x = r_{\text{imp}} \cos \theta \quad y = r_{\text{imp}} \sin \theta \quad z = z$$



- θ : circumferential coordinate.
- z : longitudinal coordinate.
- Δr : radial deviation from mean surface.
- The node topology can remain unchanged while coordinates are modified.

WAVELENGTHS

$$\lambda_{\theta} = \pi(D - t) / m$$

$$\lambda_z = 2L / \beta$$

05

Imperfection equation

Derive the equation from two independent spatial components.

05 | Circumferential + longitudinal components

The formulation is compact but each parameter has a direct physical interpretation.

CIRCUMFERENTIAL

$$f_m(\theta) = \cos[m(\theta - \theta_0)]$$

LONGITUDINAL

$$g_\beta(z) = \cos[\beta\pi(z/L - 1/2)]$$

COMBINED FIELD

$$\Delta r(\theta, z) = A \cdot f_m(\theta) \cdot g_\beta(z)$$

A

Amplitude

m

Circumferential waves

β

Longitudinal half-waves

θ_0

Phase / orientation

06

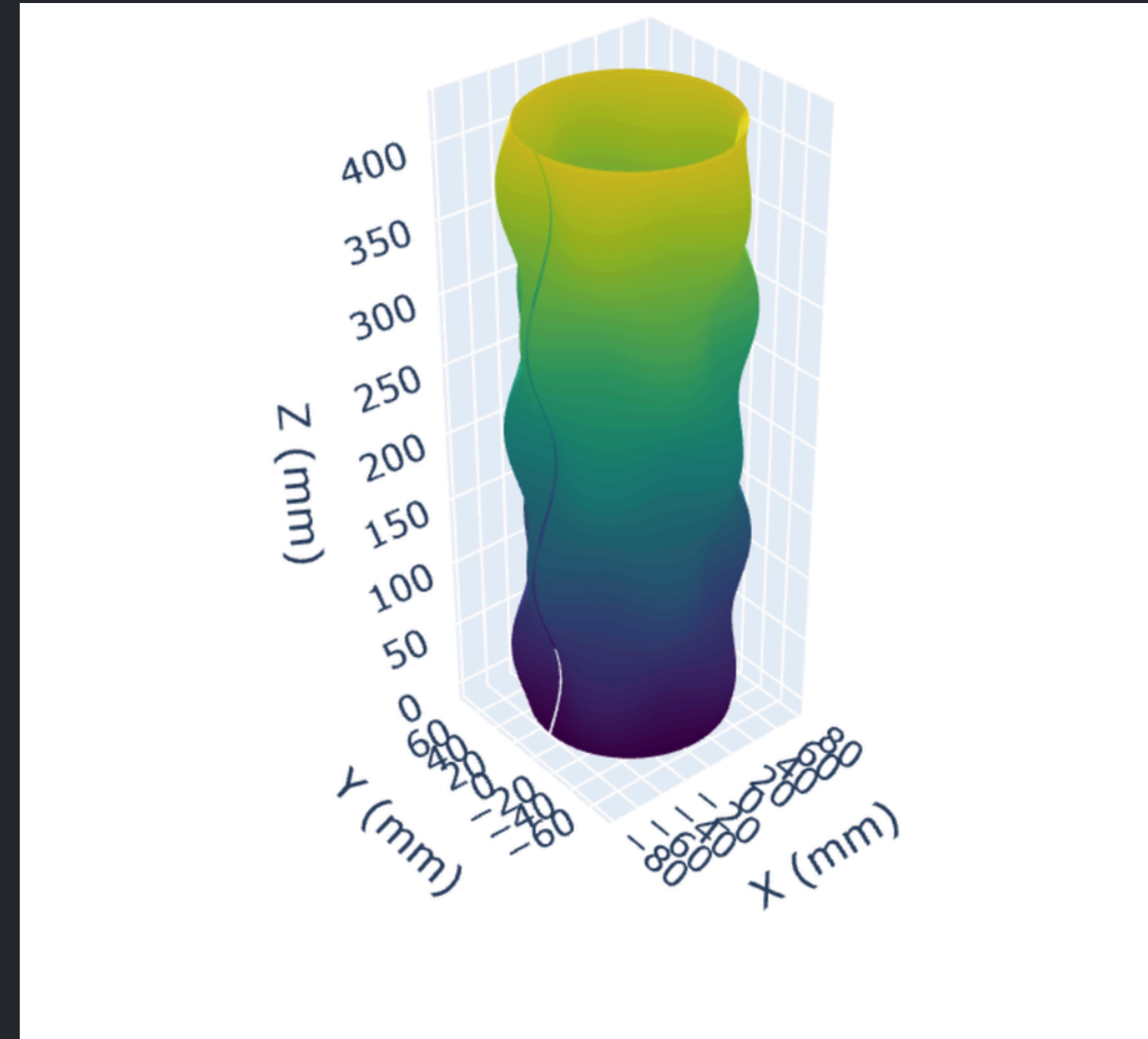
LBA-guided m and β

Use eigenmodes as physical guidance, then explore nearby imperfection families.

06 | LBA → candidate wave numbers

The workflow deliberately avoids treating the first eigenmode as universally critical.

1. Run perfect CHS LBA.
2. Identify local shell mode.
3. Count circumferential waves → m_0 .
4. Count longitudinal half-waves → β_0 .
5. Generate nearby m / β combinations.
6. Run nonlinear cases and compare response.



HERE: $m=5$ and $\beta=10$

07

Python automation

The equation becomes an executable geometry generator.

07 | From parameters to Abaqus nodes

Perfect input file → transformed node coordinates → imperfect input file.

READ

Parse instance +
node block

CALCULATE

$r, \theta, z^*, \Delta r$

TRANSFORM

$r_{new} \rightarrow x_{new}, y_{new}$

WRITE

New .inp + CSV

VERIFY

Δr extrema + λ

NODE PRE-PROCESSING

$$r = \sqrt{x^2 + y^2}, \quad \theta = \text{atan2}(y, x), \quad z^* = z - z_{\min}$$

NODE IMPERFECTION

$$\Delta r = A \cos[m(\theta - \theta_0)] \cos[\beta\pi(z^*/L - 1/2)]$$

Outputs: modified Abaqus .inp + node-level .csv for verification / visualisation



08

Verification

Geometry generation must be checked before expensive nonlinear analysis.

08 | Verification checklist

Three levels of verification make the automated pipeline auditable.

MATHEMATICAL

$\max|\Delta r| \approx A$; $\lambda\theta$ and λz correct

NODE

node count and z preserved

VISUAL

3D reconstructed surface
matches m and β

FE

low-cost Abaqus verification
run

Automated metadata: $\min \Delta r \bullet \max \Delta r \bullet \max|\Delta r| \bullet \lambda\theta \bullet \lambda z \bullet \text{node count} \bullet \text{case ID}$



09

Nonlinear FE / Riks

The imperfect geometry is analysed along the nonlinear equilibrium path.

09 | Nonlinear analysis strategy

Riks is particularly useful when the equilibrium path contains limit points or unstable portions.

MODEL

Imperfect shell geometry + material model + boundary conditions + loading.

SOLVER

Geometric/material nonlinearity with an arc-length / Riks step.

OUTPUT

Peak load or moment, displacement/rotation, equilibrium path and status.

Use one consistent response-extraction convention across the complete parametric study.

10

Sensitivity study

D/t , A/t , m and β form the core controlled variables.

10 | Parametric study

Initial project notes suggest $D/t = 30, 40, 50, 60\dots$ and $A/t = 0.10\text{--}1.00$ as starting levels.

D/t

100, 150, 200

A

0.10, 0.25, 0.50, 0.75, 1.00

m

integer circumferential wave number

β

longitudinal half-wave number

WHY DESIGN OF EXPERIMENTS MATTERS

$$N_{\text{cases}} = N_{D/t} \times N_{A/t} \times N_m \times N_{\beta}$$



11

FE dataset

Every simulation should become a traceable record.

11 | Dataset architecture

Inputs + derived variables + solver metadata + structural response + quality status.

CASE

Unique ID

GEOMETRY

D, t, L, D/t

IMPERFECTION

A/t, m, β , θ_0 , λ

MATERIAL

E, ν , f_y ...

SOLVER

Mesh, BCs, Riks

RESPONSE

Peak + curve descriptors

STATUS

Valid / failed / excluded

FILES

INP / ODB / CSV

TRACEABILITY RULE

Never store a response without the parameter tuple and solver status that generated it.

12

Automated result extraction

Close the loop from Abaqus ODB to the dataset.

12 | ODB → response pipeline

Automation should remove repetitive manual extraction.

```
# INPUT FILE NODE MODIFICATION
# =====

def modify_input(master_lines, m, beta, A):
    """
    Modify nodes in *NODE blocks.
    Only nodes lying on the CHS midsurface are changed.
    Returns:
        new input lines
        node records for CSV
        number of changed nodes
    """
    output = []
    in_node_block = False
    changed = 0
    node_records = []

    radius_tolerance = max(0.25, 0.10 * T)

    for line in master_lines:
        stripped = line.strip()

        if stripped.startswith("*"):
            in_node_block = stripped.upper().startswith("*NODE")
            output.append(line)
            continue

        if not in_node_block or not stripped:
            output.append(line)
            continue

        if stripped.startswith("***"):
            output.append(line)
            continue

        parts = [p.strip() for p in line.split(",")]

        if len(parts) < 4:
            output.append(line)
            continue
```

```
# =====
# ODB HISTORY EXTRACTION
# =====

def find_history_data(odb):
    cf3 = None
    u3 = None

    for step_name in odb.steps.keys():
        step = odb.steps[step_name]

        for region_name in step.historyRegions.keys():
            region = step.historyRegions[region_name]

            for key in region.historyOutputs.keys():
                key_upper = key.upper()

                if cf3 is None and key_upper == "CF3":
                    cf3 = region.historyOutputs[key].data

                if u3 is None and key_upper == "U3":
                    u3 = region.historyOutputs[key].data

            if cf3 is not None and u3 is not None:
                return cf3, u3

    return cf3, u3

def nearest_value(data_dict, t):
    if t in data_dict:
        return data_dict[t]
    nearest = min(data_dict.keys(), key=lambda x: abs(x - t))
    return data_dict[nearest]

def extract_results(folder, odb_path):
    odb = openOdb(path=odb_path, readOnly=True)
```

Standardise the extraction location and response definition for every case.



13

Machine learning

The FE database becomes the high-fidelity source for a future structural surrogate.

13 | ML formulation

No completed ML accuracy is claimed yet; this is the methodology for the next stage.

SURROGATE

$$\hat{y}=f(X)$$

$X=[D/t, A/t, \lambda\theta/L, \lambda z/L, m, \beta, \theta_o, \text{geometry, material, analysis}]$

Tree

RF / Extra Trees

Boosting

Gradient boosting

Kernel

SVR

Neural

MLP

- Prevent leakage in train/test split.
- Use held-out FE cases.
- Report MAE, RMSE, R^2 .
- Inspect errors by D/t , A/t , m , β .
- Flag extrapolation.



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3D-printed materials

The same architecture can later include anisotropy and manufacturing effects.

14 | Extension to additive manufacturing

Geometry is one module; material and process effects become additional modules.

Geometry

$A, m, \beta, \theta_0, D/t$

Material

anisotropy, yield, hardening

Process

build direction, defects

FE

nonlinear shell model

Data

coupled descriptors

ML

coupled surrogate

15

Limitations & quality control

Be explicit about what is modelled and what is not yet demonstrated.

15 | Limitations

This strengthens the research presentation rather than weakening it.

- Periodic cosine field $\neq$ complete measured imperfection reconstruction.
- Radial separable field does not capture every localised or random defect.
- FE results depend on mesh, material model, BCs and solver settings.
- Failed jobs must be retained as quality information.
- ML is valid only within a demonstrated training domain.
- AM extension requires independent material/process calibration.

WHAT TO ADD NEXT

Measured surface data • mesh convergence • experimental validation • automated ODB extraction • uncertainty analysis • coupled AM material modelling



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Research roadmap

A staged path keeps the project scientifically controlled.

17 | Roadmap

Build complexity only after the previous layer is verified.

PHASE 1

Equation + Python validation

PHASE 2

LBA-guided m/β

PHASE 3

Nonlinear FE / Riks

PHASE 4

Automated ODB extraction

PHASE 5

Large FE dataset

PHASE 6

ML surrogate

PHASE 7

Measured imperfections

PHASE 8

3D-printed materials +
validation

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Conclusion

A reproducible computational framework is the central outcome.

INTEGRATED PIPELINE

Perfect CHS $\rightarrow$ LBA $\rightarrow \Delta r(\theta, z) \rightarrow$ Python $\rightarrow$ Abaqus $\rightarrow$ FE data $\rightarrow$ ML $\rightarrow$ AM

The project combines a physically interpretable imperfection model with reproducible FE automation and a scalable data pathway.

Thank you